Qns	Answer	Marks
2(a)	g and r are constants so a is proportional to x the negative sign shows that a and x are in opposite directions/ a is always directed towards the equilibrium	B1 B1
2(b)	since $a = -\frac{gx}{r}$ and $a = -\omega^2 x$ so $\omega^2 = \frac{g}{r}$ $\omega^2 = \frac{9.81}{0.28}$ $\omega = 5.9 \text{ rad s}^{-1}$ *(5.92)*	 M1 A1
2(c)(i)	Negative sine graph max a = $\omega^2 x_o$ = 5.92 (0.05) = 1.7 m s^{-2}	B1 B1
(c)(ii)	since this is a light damping oscillation, period remain the same as before and remains constant over time Amplitude consistently decreases and first peak must be less than 5.0cm	B1 B1
(c)(iii)	Loss in energy $= \frac{1}{2} m\omega^2 {x_{oi}}^2 - \frac{1}{2} m\omega^2 {x_{of}}^2$ $= \frac{1}{2} \times 0.090 \times 5.9^2 \times 0.050^2 - \frac{1}{2} \times 0.090 \times 5.9^2 \times 0.025^2$ M1 $= 0.294 = 0.29 \text{ J}$ A1	 M1 A1